nsdi '26

MAE: More Adaptive Video Encoder for Consistent Low Latency in High-Quality Real Time Communication

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Real Time Communication - One of the key uses on the Internet today

Video Conference

Cloud Gaming / PC VR

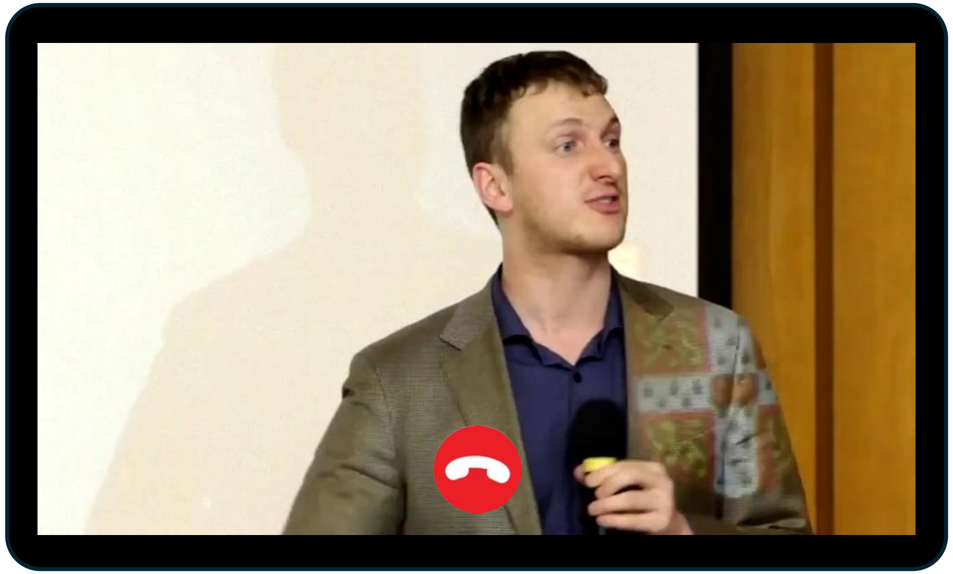
Remote Surgery

The global video conferencing market reached **US$11.65 billion** in 2024, alongside the global cloud gaming market at **US$2.27 billion**.*

* Data from https://www.grandviewresearch.com/industry-analysis

Maintaining consistent low latency is important to user experience

Have you experience this?



99.9th percentile tail latency

↓

Every 1000 frames experience one stall

↓

Every 30 seconds!

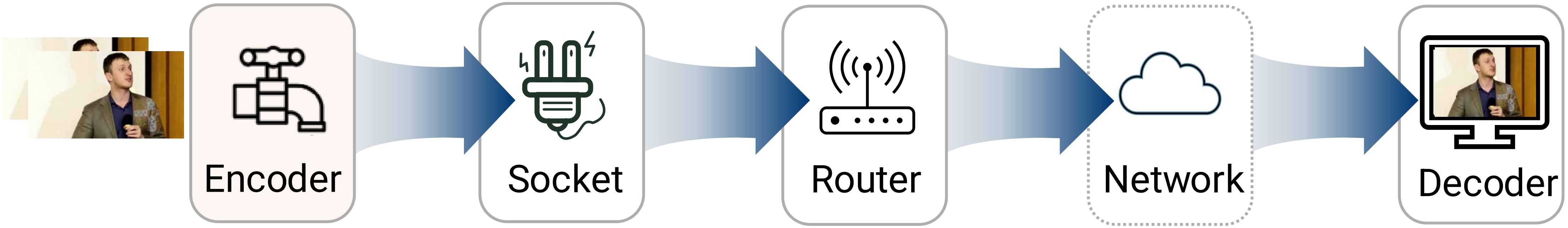
😠 Current **stall** experiences are **still far from satisfactory**.

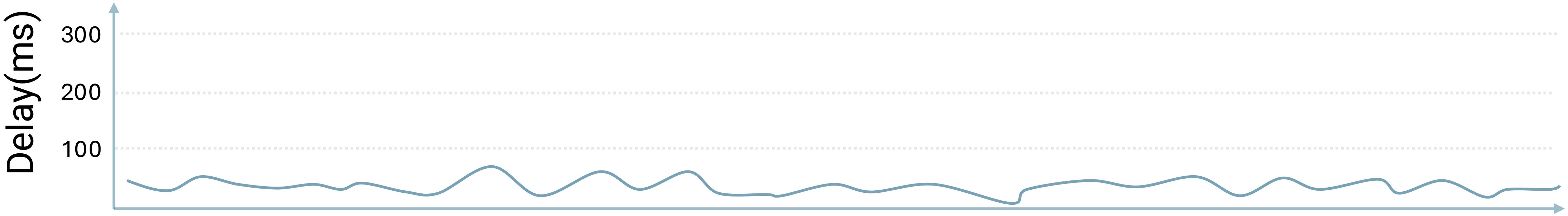
Large-scale live streaming data shows wireless tail latency hits **~200ms** at the **99.9th** percentile.*

* Data from AUGUR, NSDI 2024

Why existing solution failed?

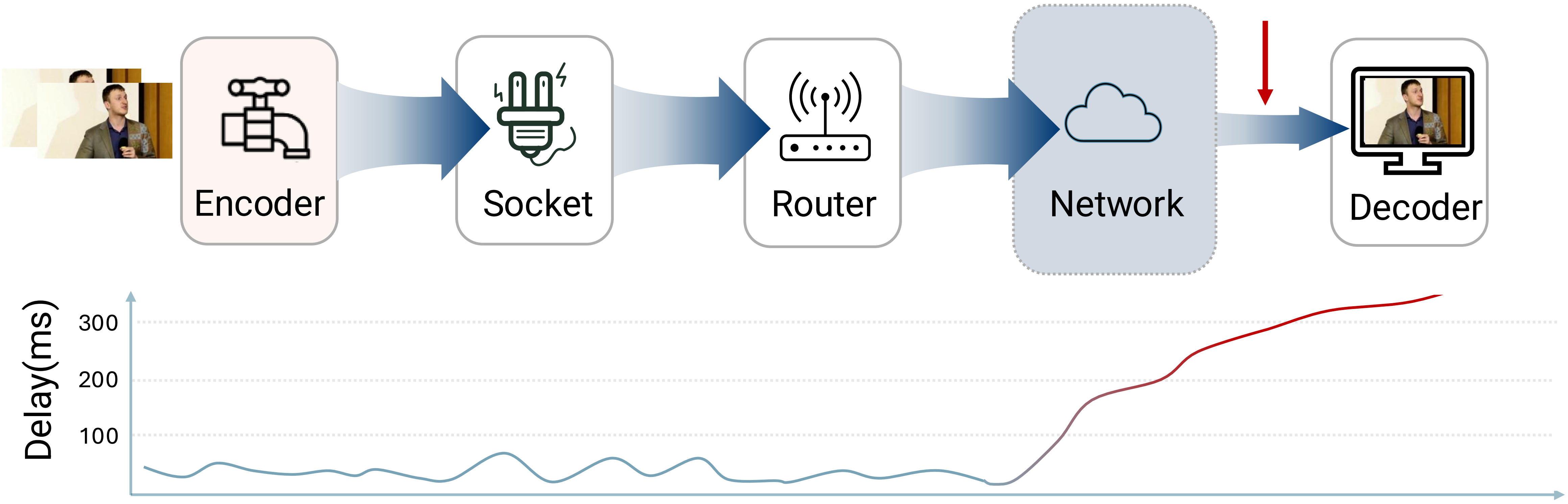
This shows a simplified real-time communication pipeline, which resembles a **water pipeline**.





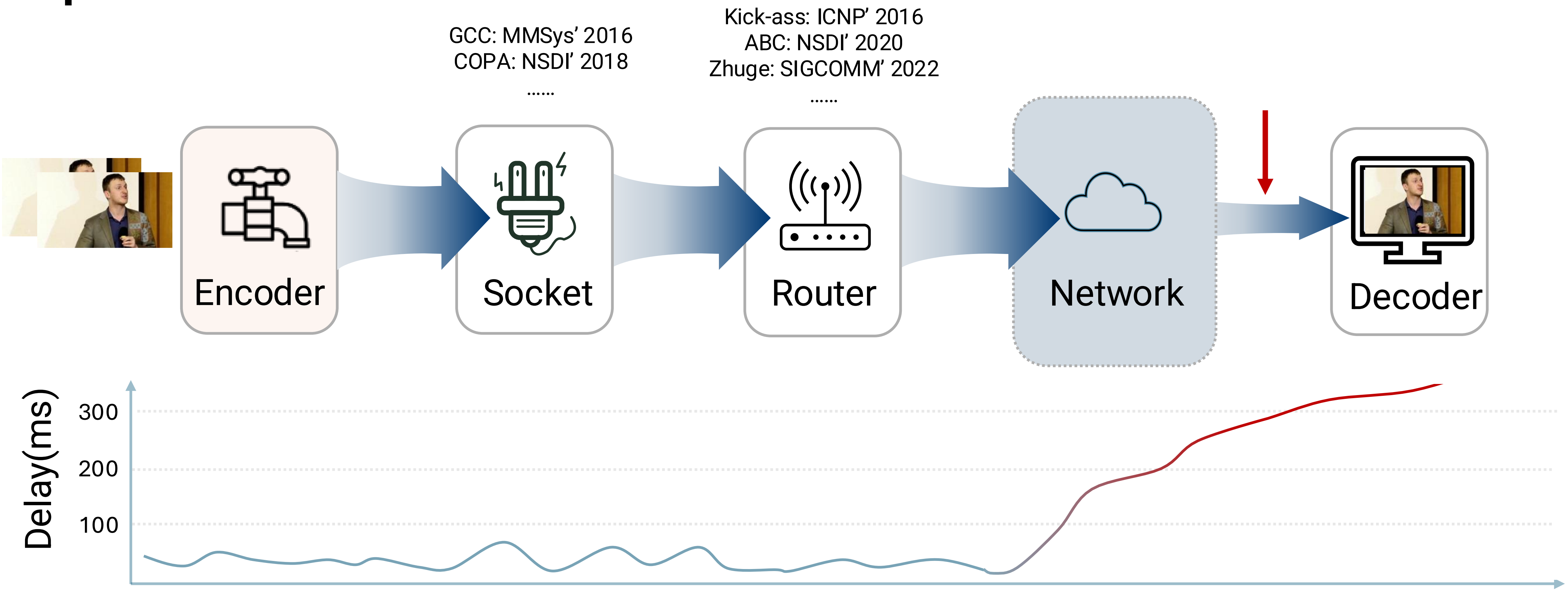
Why existing solution failed?

However, unlike a water pipeline, network bandwidth **fluctuates** dynamically over time.



Why existing solution failed?

Previous work simply **shifts** the waiting queue from **downstream to upstream**.



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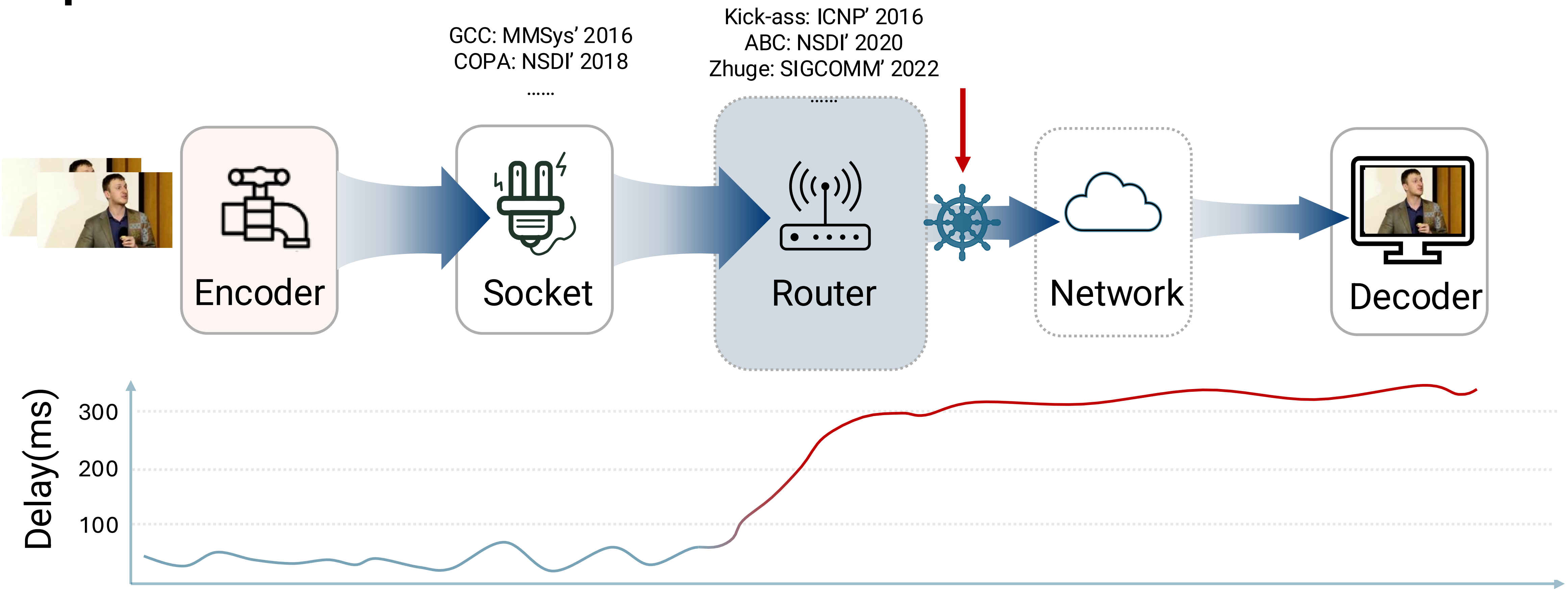
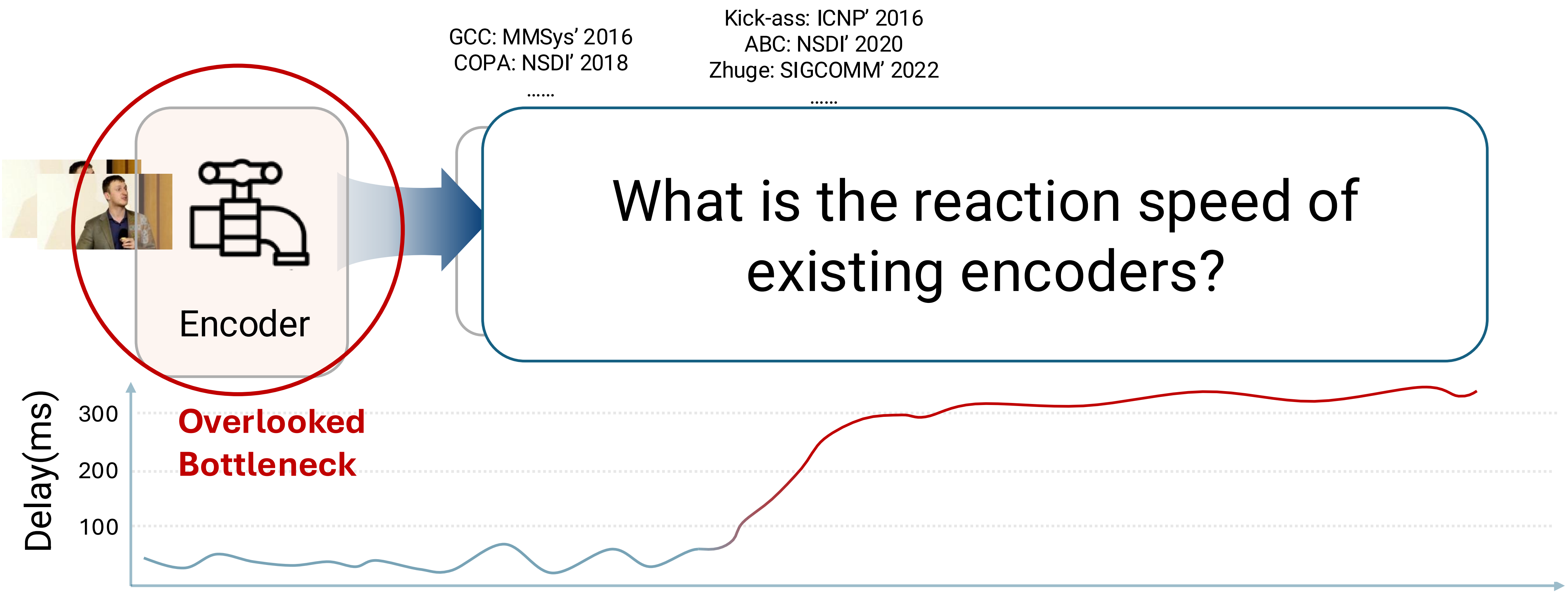


Why existing solution failed?

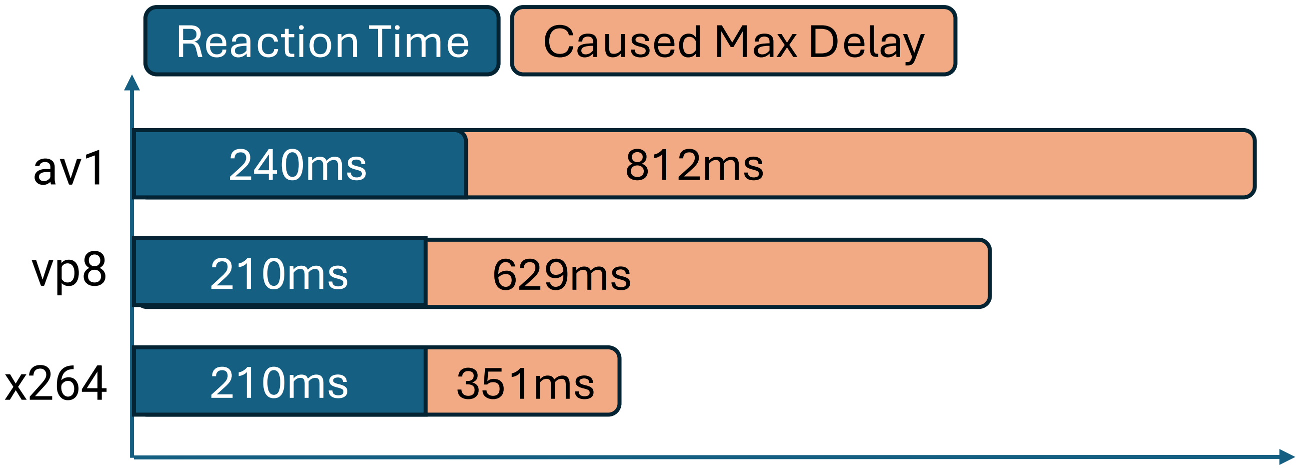
The True Bottleneck: A **Slow-Reacting Source** - Encoder



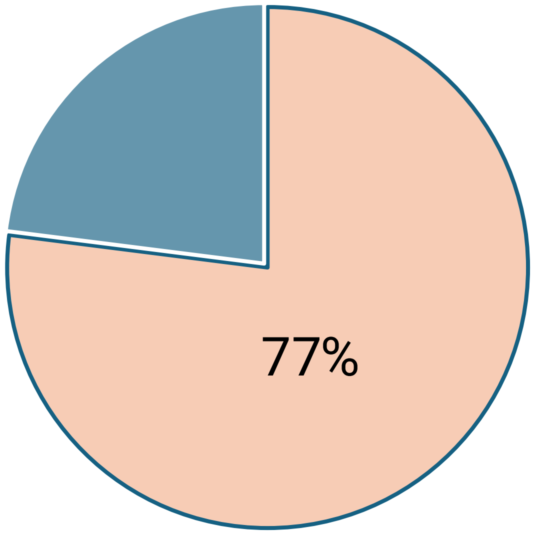
Why existing solution failed?

The True Bottleneck: A **Slow-Reacting Source** - Encoder

What is the reaction speed of existing encoders?



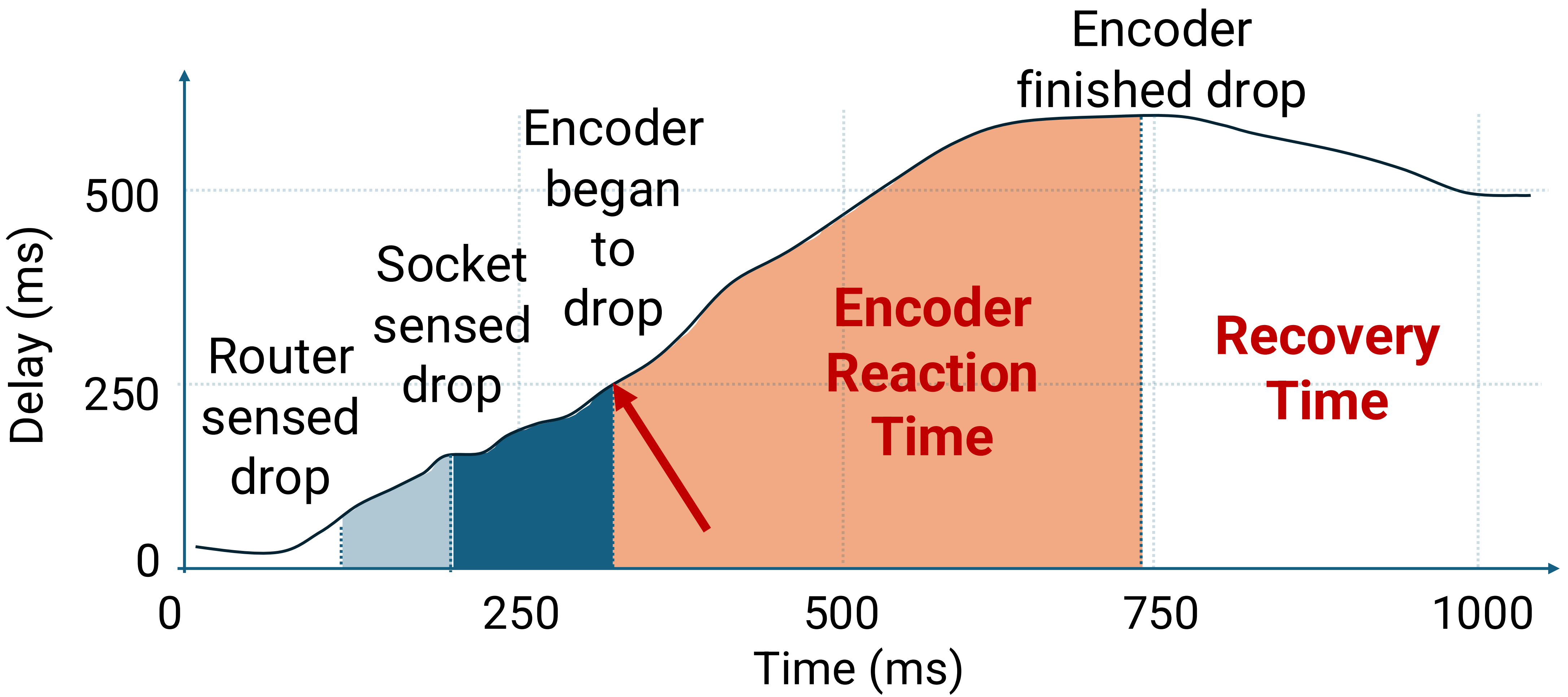
Reaction Speed Under 10x Bandwidth Drop



Our experiment data shows that **77%** reaction time comes primarily from the encoder's reaction time.

Why existing solution failed?

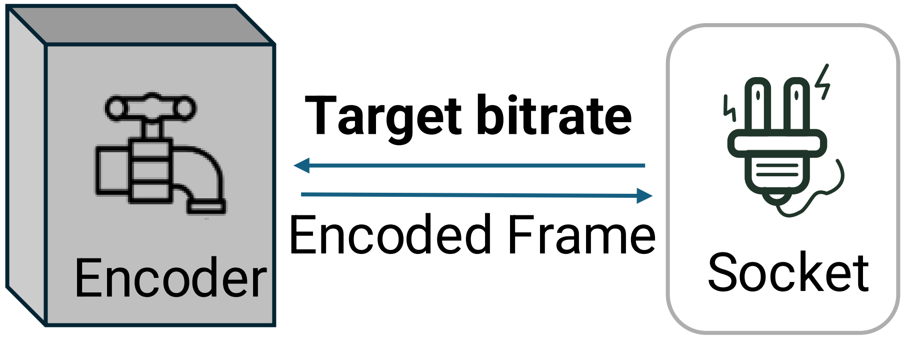
Dive into a detailed bandwidth drop adaptation process.



Detailed reaction time when bandwidth drops from 10Mbps to 1Mbps.

Why current encoders react slowly?

Encoder lacks decision information.



Treated as a black box.

Why current encoders react slowly?

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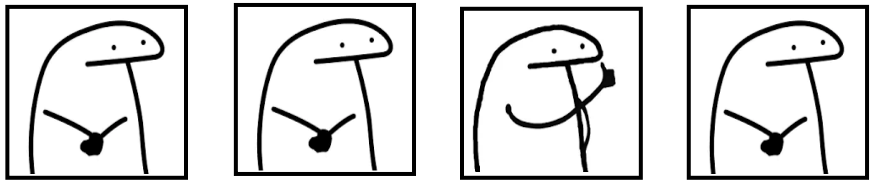
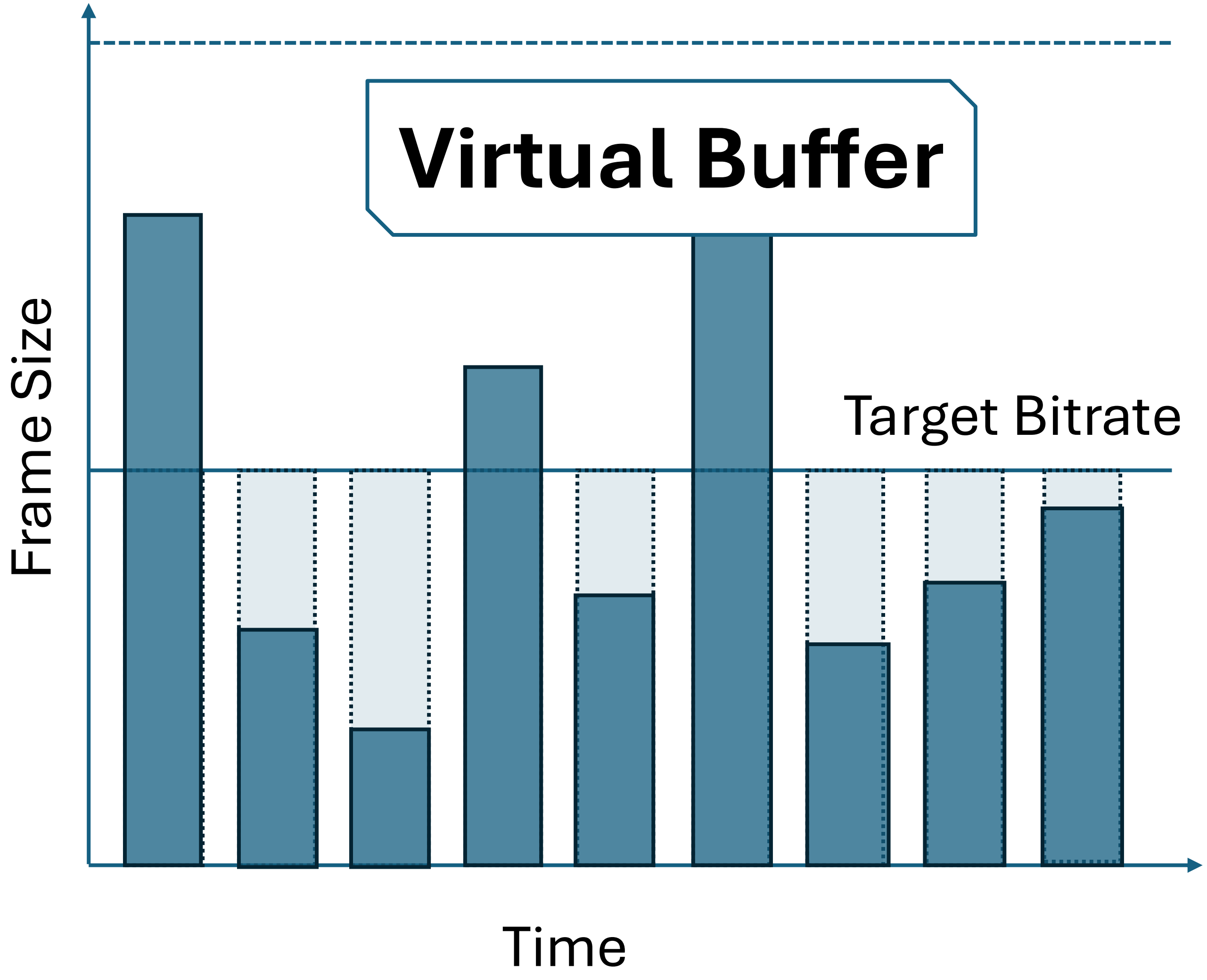
How to control the overshoot?



But content is fluctuated.



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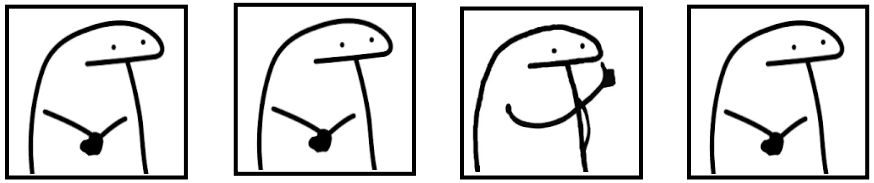
How does the virtual buffer work?



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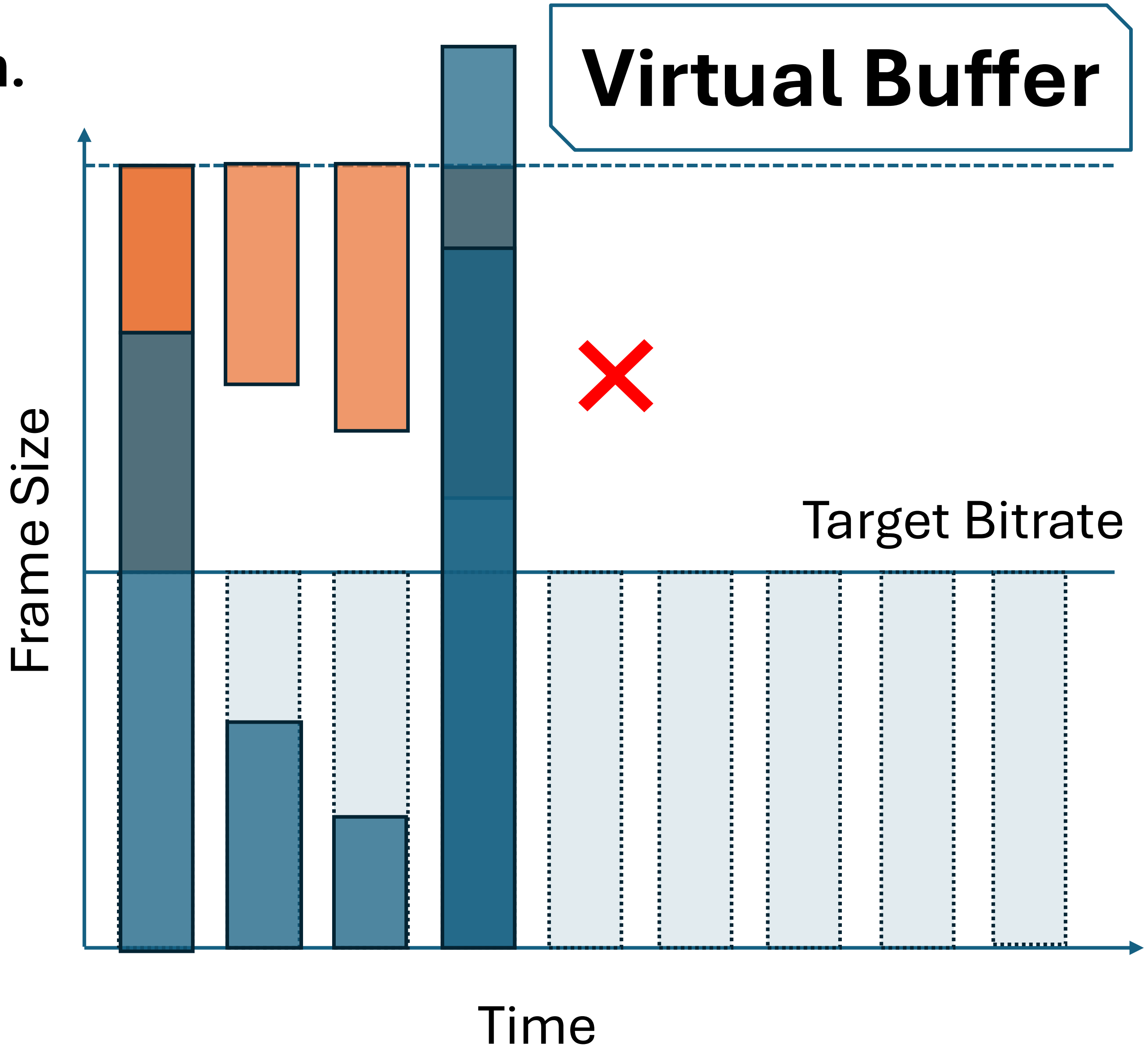

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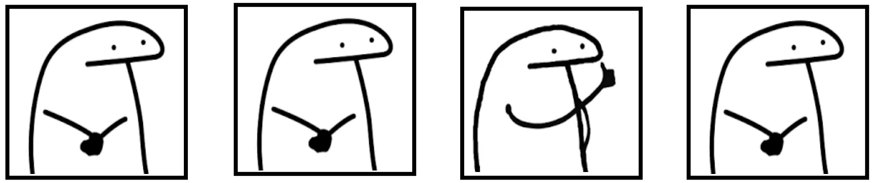
What will happen when bandwidth drops?



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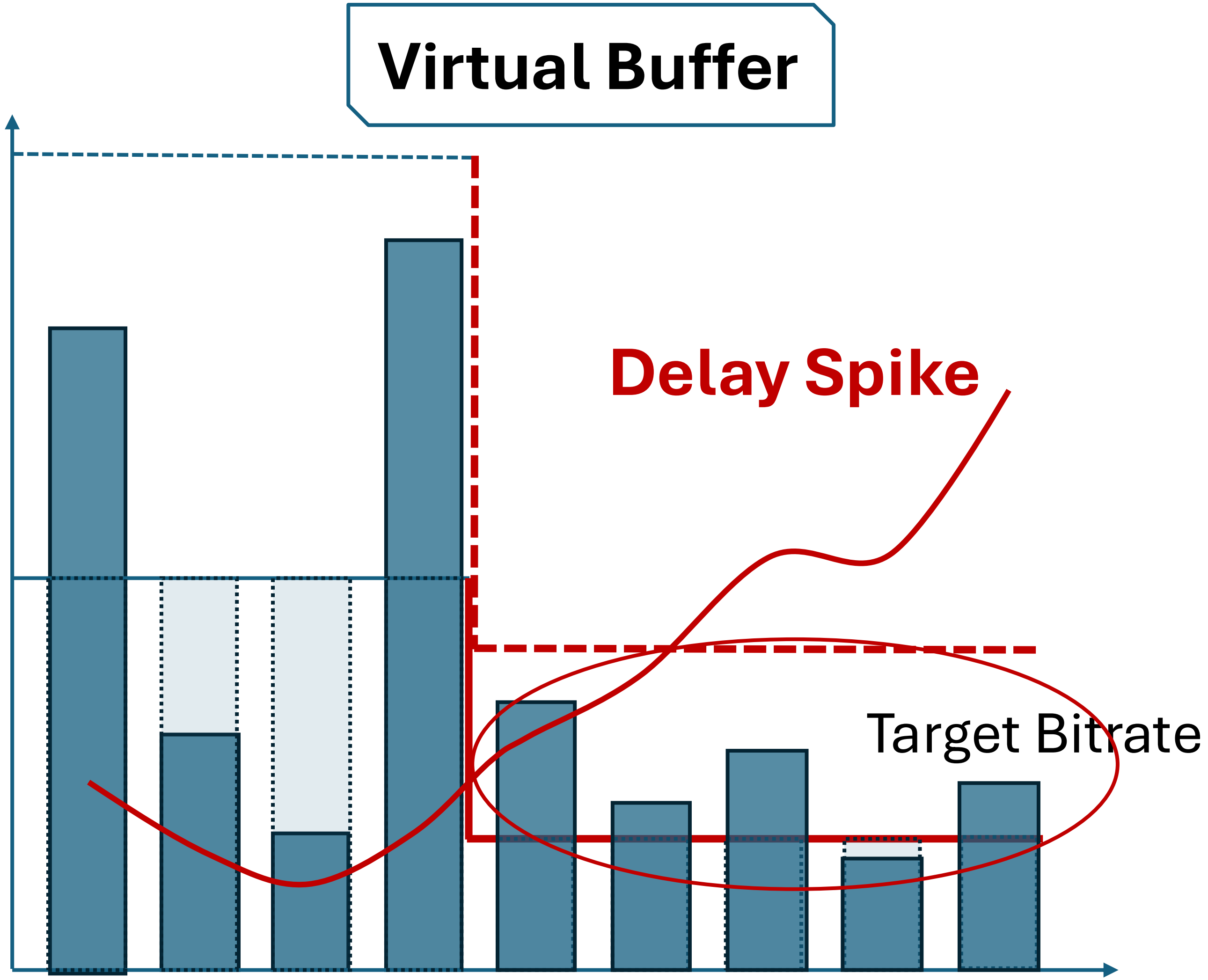

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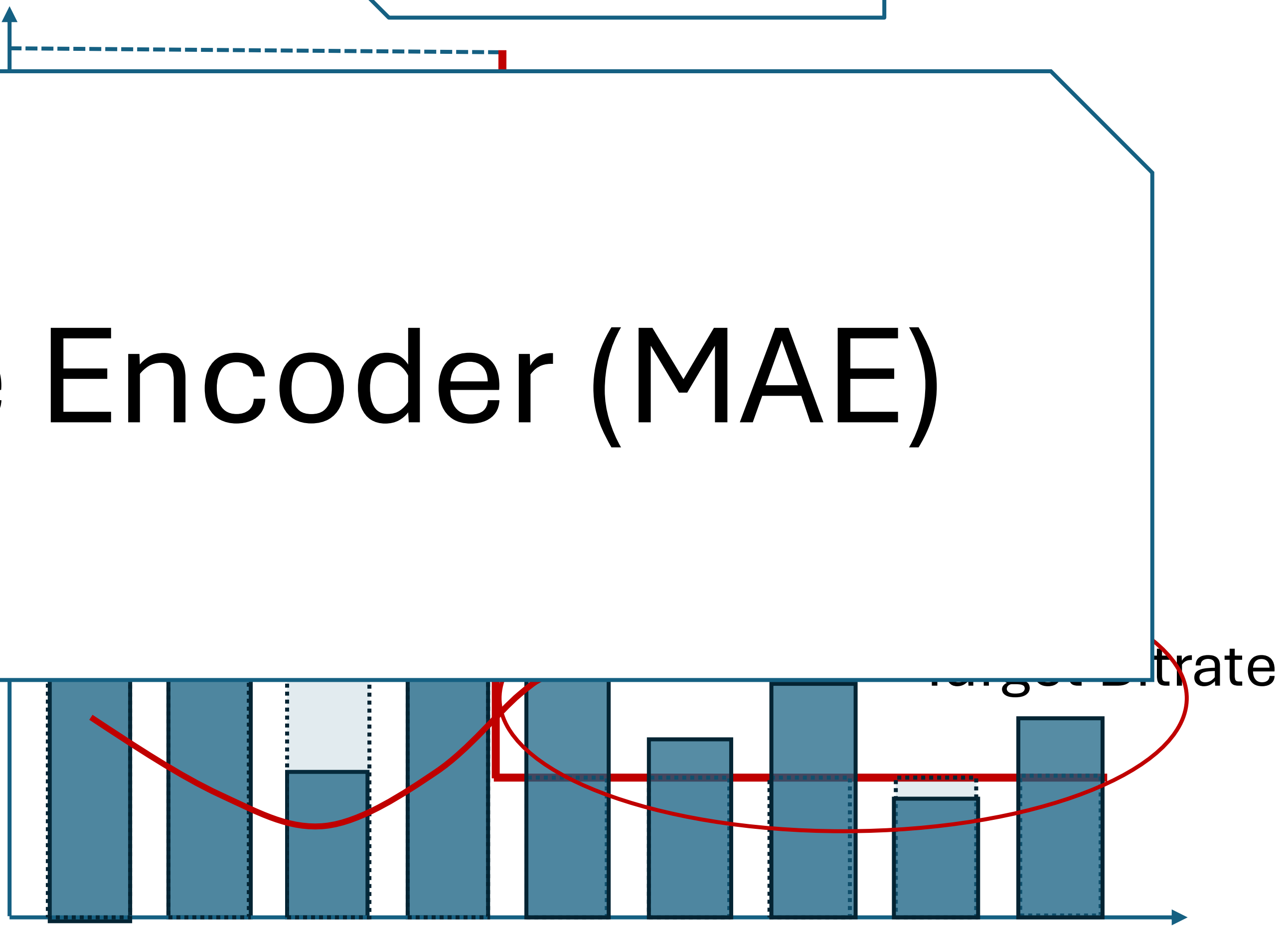
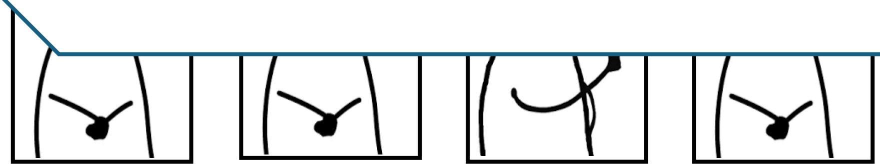
Virtual Buffer

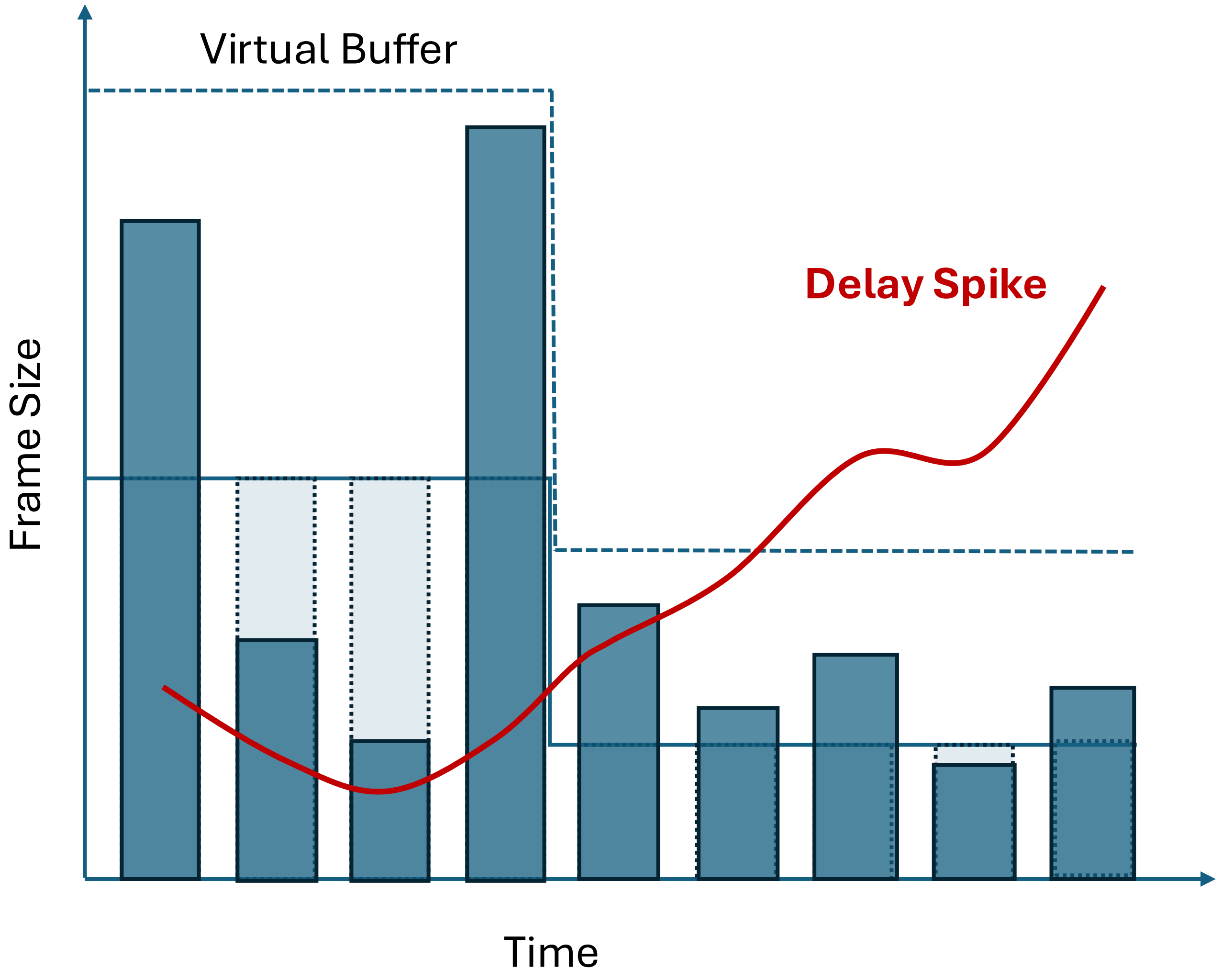


More Adaptive Encoder (MAE)



But content is fluctuated.

MAE 1: Dynamic Virtual Buffer Sizing



	Always large buffer
Quality	Good
Latency	Bad

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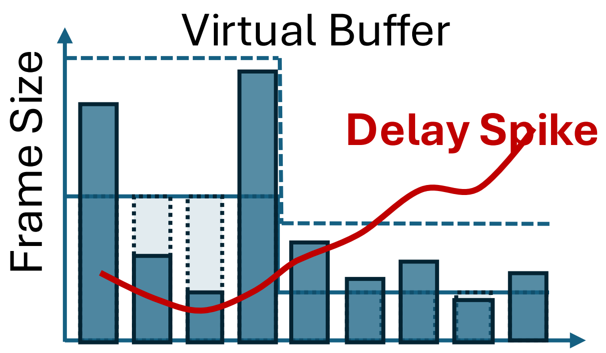
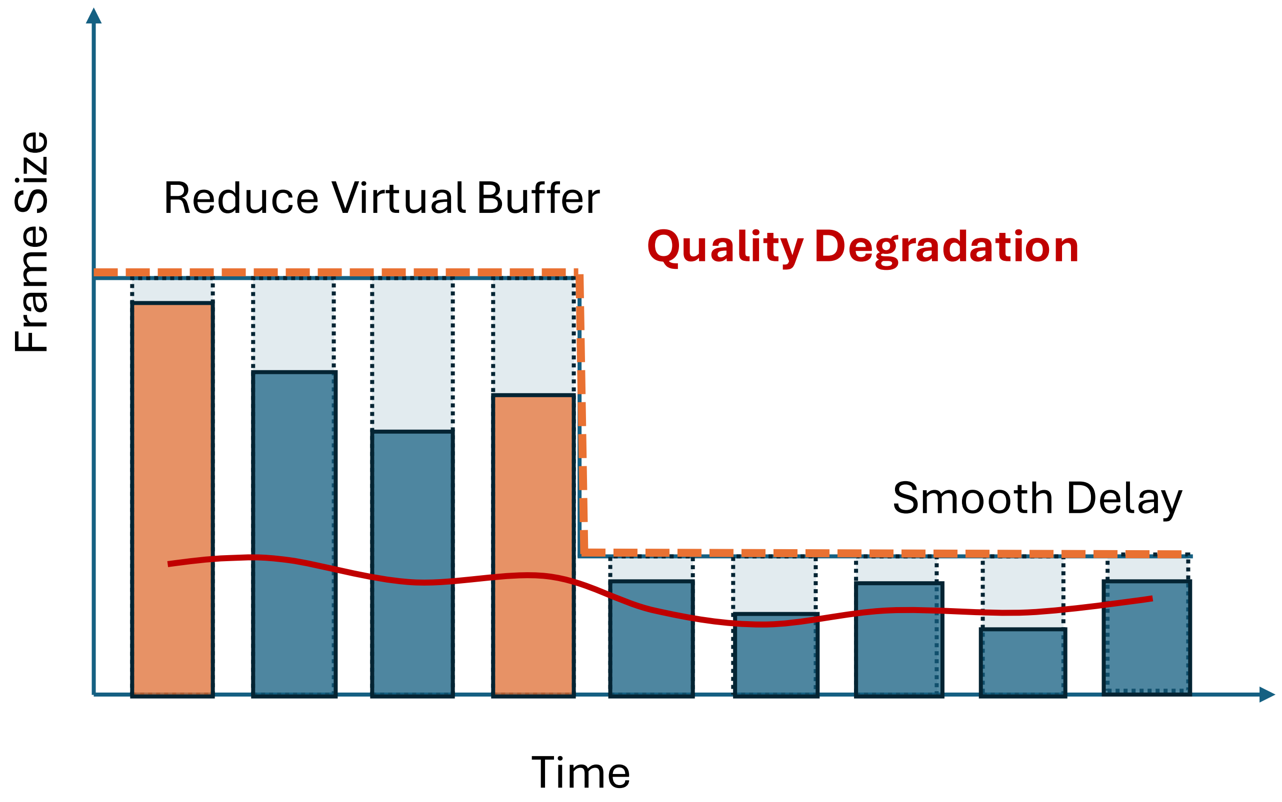


	Always large buffer	Always small buffer
Quality	**Good**	**Bad**
Latency	**Bad**	**Good**



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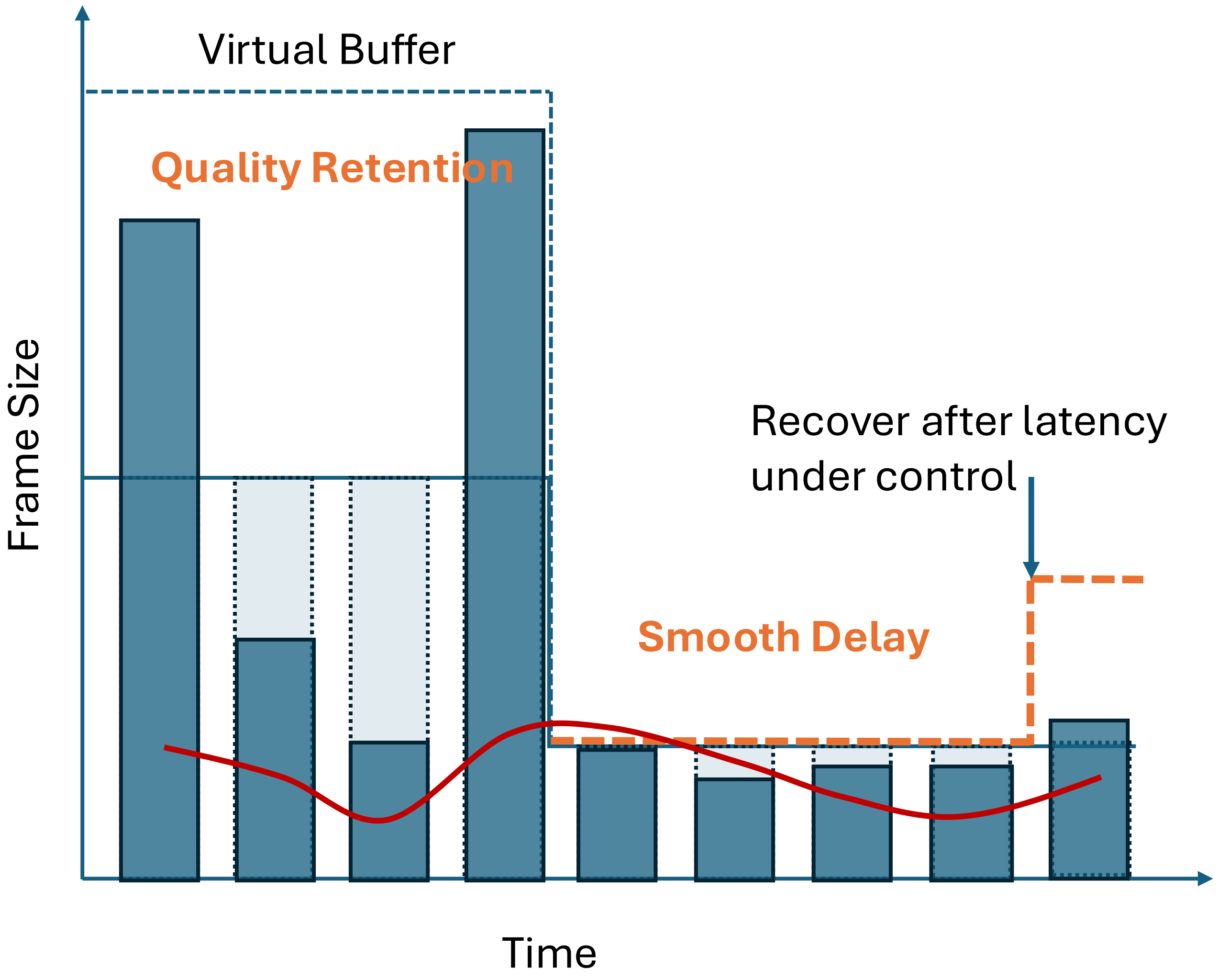

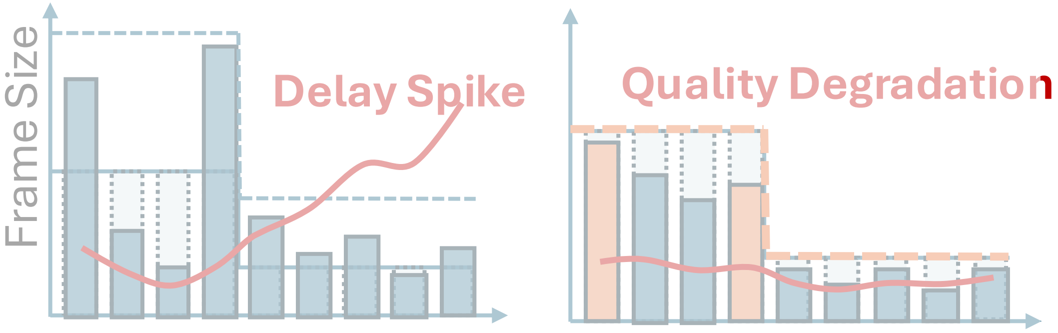

	Always large buffer	Always small buffer
Quality	Good	Bad
Latency	Bad	Good

MAE **adapts** it for different network condition.

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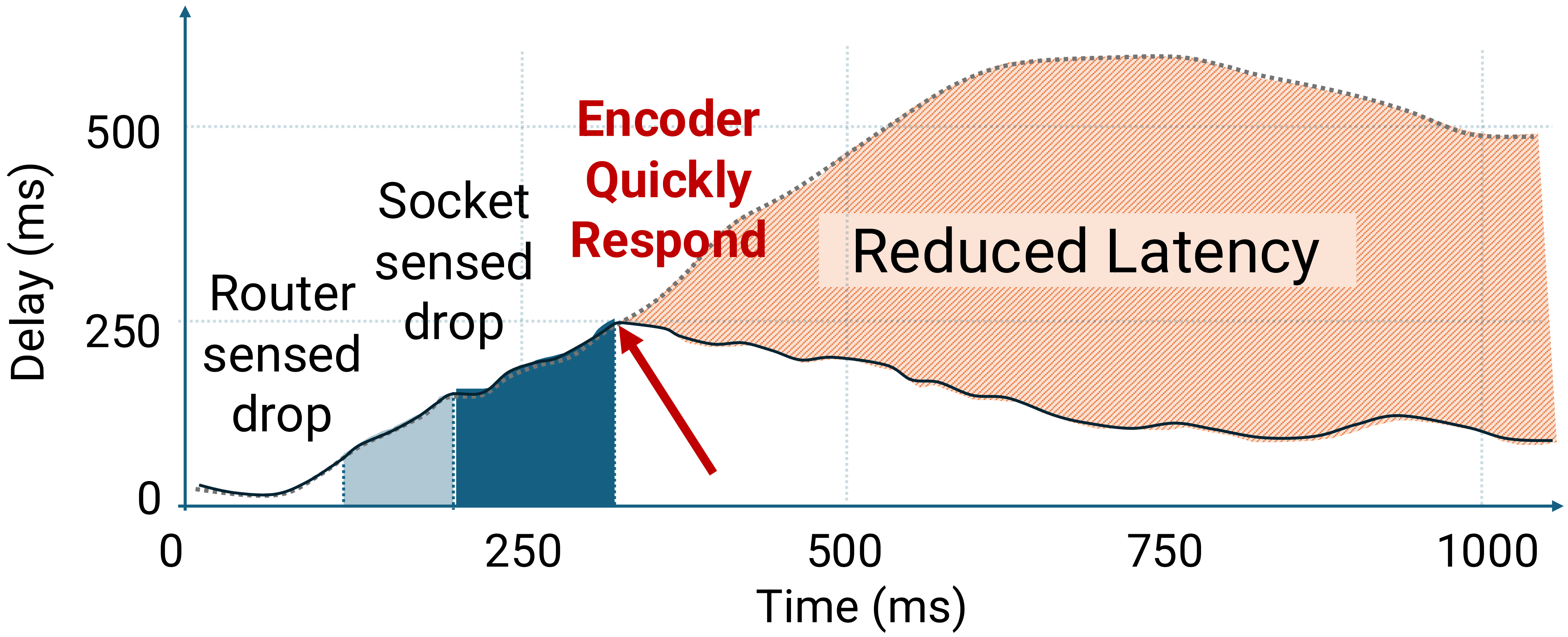
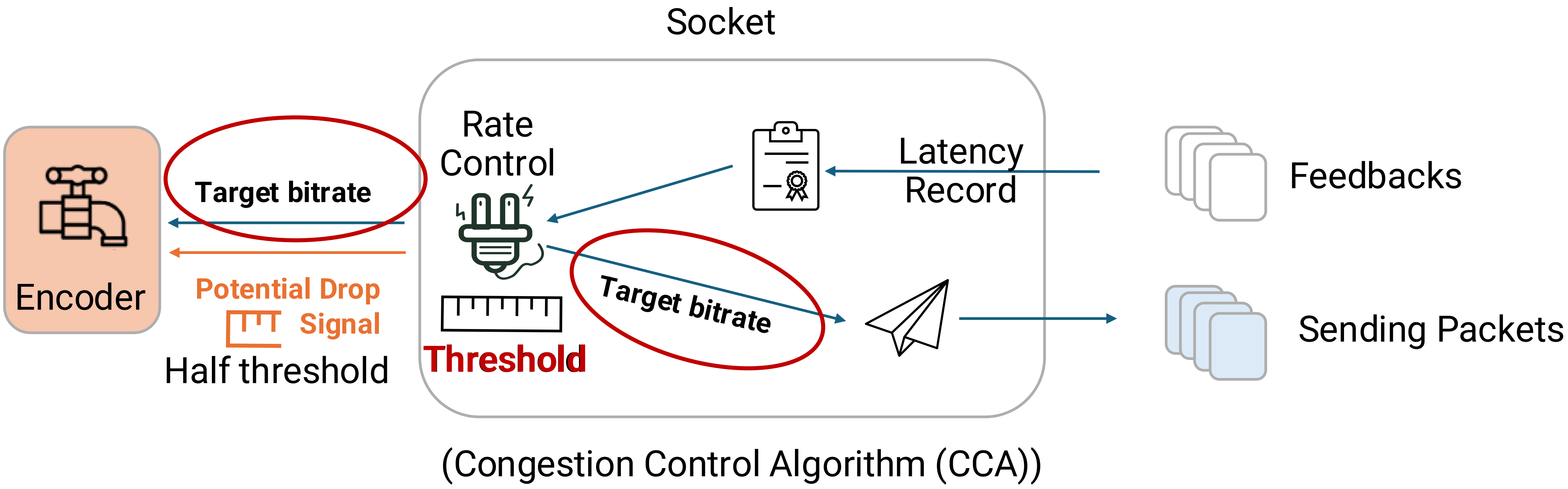


Detailed reaction time when bandwidth drops from 10Mbps to 1Mbps.

MAE 2: Pre-reacting based on CCA internal state probing



Why can't we directly reduce the original threshold?

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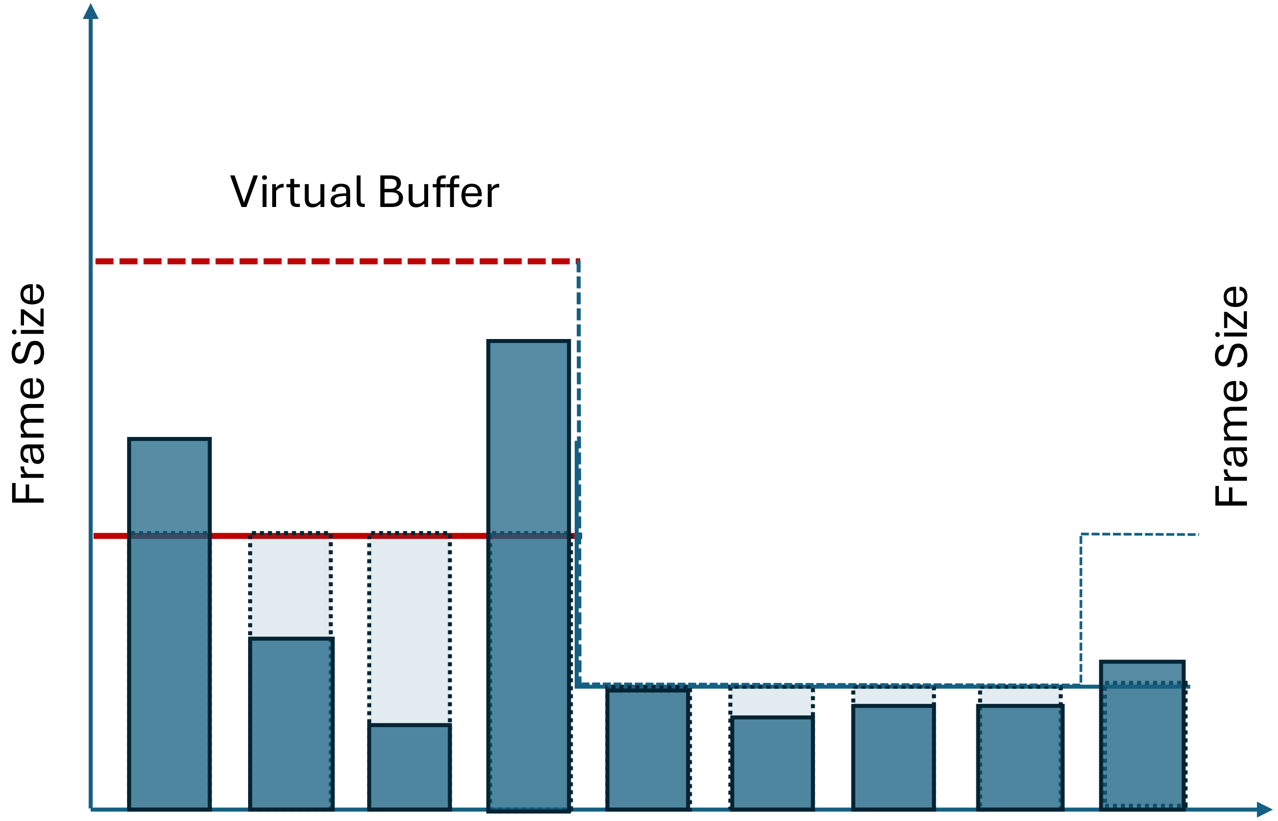
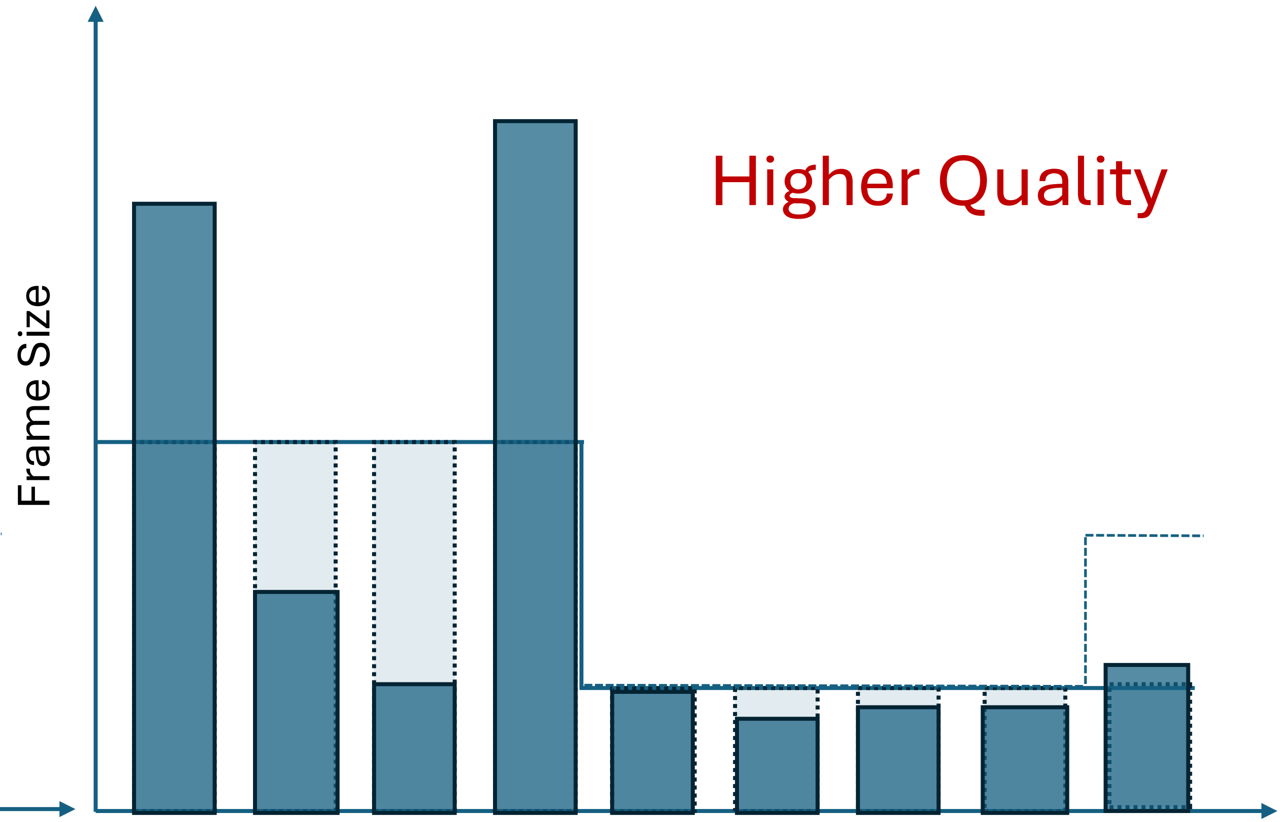


Directly reduce original threshold



Only encoder adapts virtual buffer

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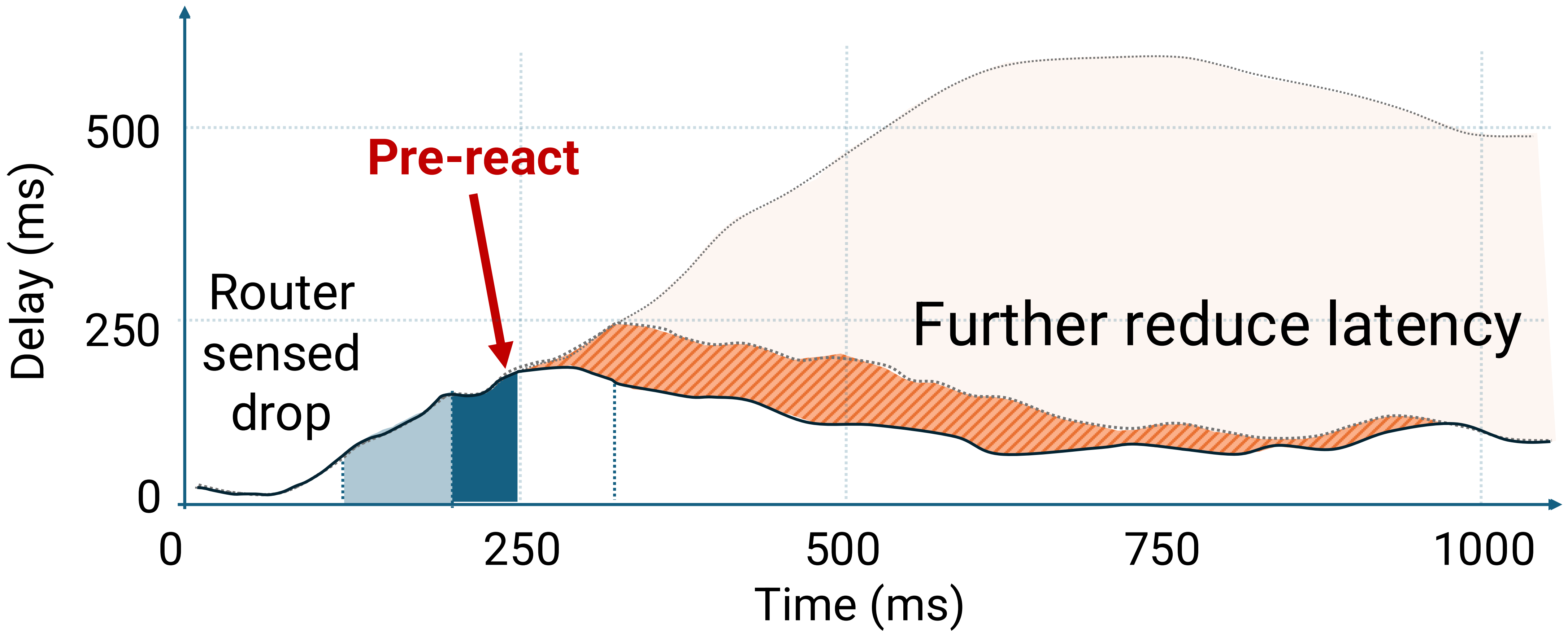
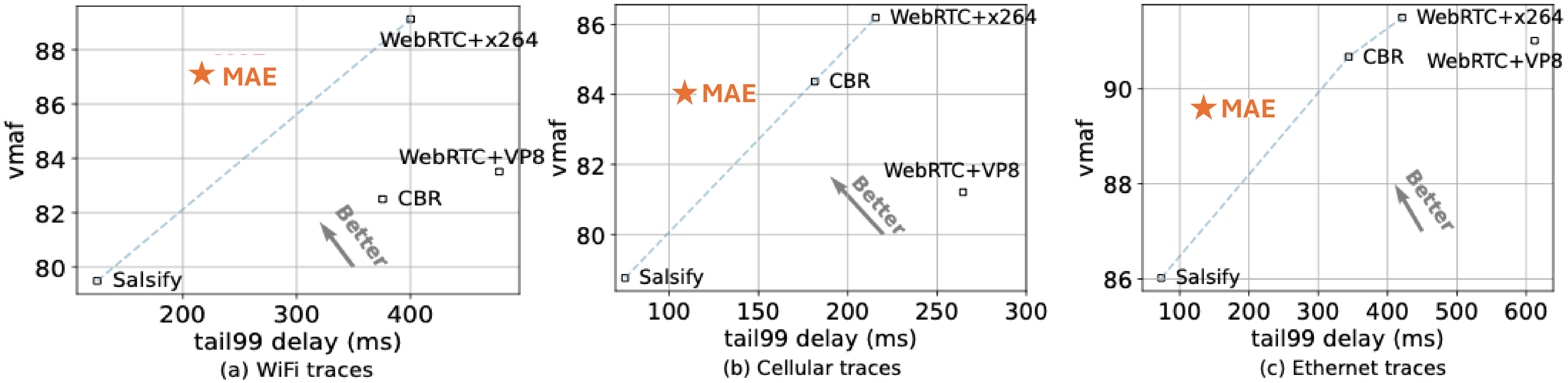


Detailed reaction time when bandwidth drops from 10Mbps to 1Mbps.

Empirical performance across diverse networks



Overall Stalls

86.2% stall reduction.

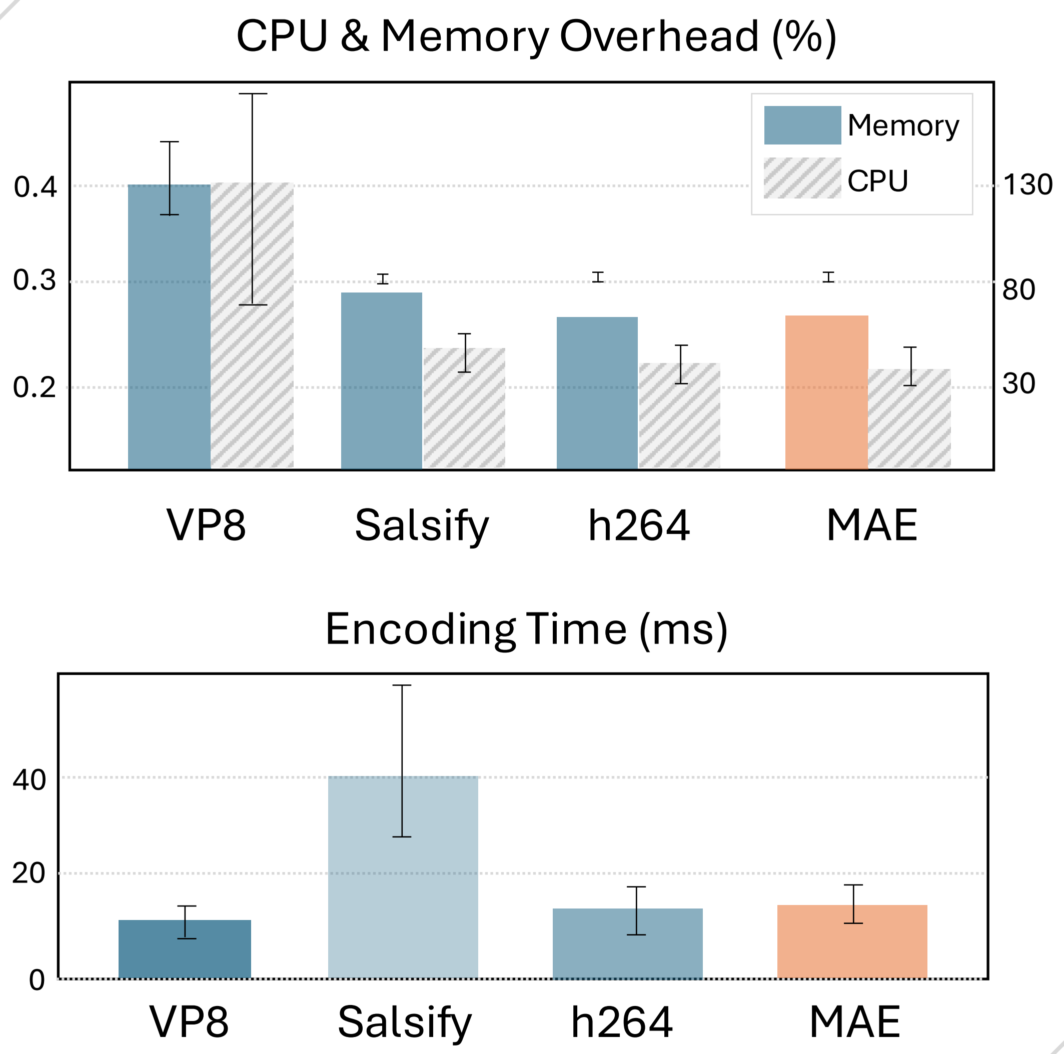
Global stall ratio drops massively from 3.8% to 0.69%.

High-Motion Quality Retention

11%.

In high-motion gaming content, MAE outperforms Salsify's VMAF by 11% by intelligently allocating bits during stable bandwidth periods.

Production-Ready: Zero System Overhead



Architectural Generality

Bandwidth Utilization

At **most 4%** reduction, barely impacts video quality.

CCA Agnostic Integration

Extended to Copa CCA, reducing **47.5%** stall ratio.

Takeaways

1. The **slow-response encoder** is the *core bottleneck* behind existing playback stalls.
2. The encoder possesses many **controllable potential**; treating it as a black box fails to leverage its inherent capability.
3. Current congestion control algorithms have the potential to **pre-react** before actual bandwidth drop signals emerge.
4. Our MAE framework requires **no modifications** to existing encoders, congestion control algorithms, or network protocols. It reduces stall rate by **86.2%** while preserving identical quality.

Thank you!

Project Website

https://github.com/hkust-spark/sparkrtc

Personal Website

https://huameng15.github.io